

15.4 Orthogonal contrasts

Once the F test of [Section 15.3](#) says that the treatment component of the mean is not zero, the next step is to split that component into interpretable pieces. A complete set of orthogonal contrasts chooses an orthogonal basis of the treatment space, one direction per question, and splits the between-groups sum of squares along it. [Chapter 8](#) proved the arithmetic ([Proposition 8.7.2](#)) and [Chapter 9](#) placed it in the theory of single-degree-of-freedom decompositions ([Theorem 9.4.4](#)). This section adds the distribution theory, builds orthogonal sets, and analyses the shape of the nitrogen response.

15.4.1 The decomposition

Contrasts $\mathbf{c}$ and $\mathbf{d}$ are orthogonal for the design if $\sum_k c_k d_k / n_k = 0$ ([Definition 8.7.1](#)), which says exactly that $\mathbf{u}_\mathbf{c} \perp \mathbf{u}_\mathbf{d}$ ([Proposition 8.7.2\(c\)](#)). Since $\dim \mathcal{T} = g - 1$, a set of $g - 1$ pairwise orthogonal contrasts is as large as possible; we call it **complete**.

Theorem 15.4.1. Orthogonal contrasts decompose the treatment sum of squares

In the one-way model, let $\mathbf{c}_1, \dots, \mathbf{c}_{g-1}$ be contrasts that are pairwise orthogonal for the design,

$$\sum_{k=1}^g \frac{c_{ik} c_{lk}}{n_k} = 0 \quad (i \neq l), \quad (15.4.1)$$

and write $\psi_i = \sum_k c_{ik} \mu_k$ and $F_i = \text{SS}(\mathbf{c}_i) / s^2$.

- (a) The vectors $\mathbf{u}_{\mathbf{c}_1}, \dots, \mathbf{u}_{\mathbf{c}_{g-1}}$ form an orthogonal basis of the treatment space $\mathcal{T}$, and

$$\text{SSB} = \sum_{i=1}^{g-1} \text{SS}(\mathbf{c}_i).$$

- (b) In the normal one-way model, $\text{SS}(\mathbf{c}_1), \dots, \text{SS}(\mathbf{c}_{g-1})$ and SSE are mutually independent, and $\text{SS}(\mathbf{c}_i) / \sigma^2 \sim \chi^2(1, \gamma_i)$ with $\gamma_i = \psi_i^2 / (\sigma^2 v_{\mathbf{c}_i})$. The noncentralities add up to that of the F test: $\sum_i \gamma_i = \gamma$ of [Eq. 15.3.3](#).
- (c) The F statistic is the average of the single-degree-of-freedom statistics: $F = (g - 1)^{-1} \sum_i F_i$.
- (d) In a balanced layout, [Eq. 15.4.1](#) reduces to $\sum_k c_{ik} c_{lk} = 0$.
- (e) Conversely, every orthogonal basis $\mathbf{u}_1, \dots, \mathbf{u}_{g-1}$ of $\mathcal{T}$ arises in this way: if $\mathbf{u}_i$ takes the value v_{ik} on group k , then $c_{ik} = n_k v_{ik}$ defines a complete set of orthogonal contrasts with $\mathbf{u}_{\mathbf{c}_i} = \mathbf{u}_i$.

Proof. Part (a) is [Proposition 8.7.2\(d\)](#), whose proof shows that the $\mathbf{u}_{\mathbf{c}_i}$ are nonzero, pairwise orthogonal vectors of $\mathcal{T}$, hence a basis, and that the projections $\mathbf{P}_i$ onto the lines they span sum to $\mathbf{P}_\mathcal{T}$. (b) The $\mathbf{P}_i$ are mutually orthogonal projections of rank one, and each is orthogonal to $\mathbf{I} - \mathbf{M}$ because its range lies in $\mathcal{C}(\mathbf{Z})$. By [Theorem 4.4.4](#) the sums of squares $\|\mathbf{P}_i \mathbf{Y}\|^2 = \text{SS}(\mathbf{c}_i)$ ([Proposition 15.2.1\(c\)](#)) and SSE are independent, with noncentralities $\|\mathbf{P}_i \boldsymbol{\theta}\|^2 / \sigma^2$. By the computation in the proof of [Proposition 15.2.2](#), $\|\mathbf{P}_i \boldsymbol{\theta}\|^2 = (\mathbf{u}_{\mathbf{c}_i}^\top \boldsymbol{\theta})^2 / \|\mathbf{u}_{\mathbf{c}_i}\|^2 = \psi_i^2 / v_{\mathbf{c}_i}$. Since $\sum_i \mathbf{P}_i = \mathbf{P}_\mathcal{T}$ and the $\mathbf{P}_i \boldsymbol{\theta}$ are orthogonal, $\sum_i \|\mathbf{P}_i \boldsymbol{\theta}\|^2 = \|\mathbf{P}_\mathcal{T} \boldsymbol{\theta}\|^2$, which is $\sigma^2 \gamma$ by

Theorem 15.3.1. (c) Divide (a) by $(g-1)s^2$. (d) Put $n_k = m$. (e) In [Section 15.2](#) we saw that each $\mathbf{u} \in \mathcal{T}$ equals $\mathbf{u}_c$ for the contrast $c_k = n_k v_k$, and by Proposition 8.7.2(c), $\sum_k c_{ik} c_{lk} / n_k = \mathbf{u}_{c_i}^\top \mathbf{u}_{c_l}$, which is zero for $i \neq l$. ■

By part (e) there are infinitely many complete orthogonal sets, one for each orthogonal basis of $\mathcal{T}$, and by part (a) all of them split the same total. The individual pieces depend on the basis ([Exercise \(C1\)](#)): choosing the basis is choosing the questions, and the design of the study, not the mathematics, should make that choice.

Warning. Orthogonality is a convenience, not a requirement. Planned contrasts that are not orthogonal are valid, each with the test and interval of Proposition 15.2.1. What is lost is additivity: their sums of squares need not total SSB, and their estimates are correlated. No contrast should be replaced by a less relevant one merely to achieve orthogonality.

15.4.2 Building orthogonal sets

Three constructions cover most designed experiments.

Splitting the treatments in stages. Many questions form a hierarchy: control against treatments, then old treatments against new, then one new formulation against another. Each splits a set of levels into two parts, and the comparisons are orthogonal for any group sizes provided each part is summarized by the mean of all its observations.

Proposition 15.4.2. Contrasts from successive splits

For a set S of levels split into disjoint nonempty parts A and B , let $N_A = \sum_{k \in A} n_k$ and $N_B = \sum_{k \in B} n_k$, and define the contrast

$$c_k^{(S)} = \begin{cases} n_k/N_A, & k \in A, \\ -n_k/N_B, & k \in B, \\ 0, & k \notin S. \end{cases}$$

Its estimate is $\bar{y}_A - \bar{y}_B$, the difference between the means of all observations in A and in B , and

$$\text{SS}(\mathbf{c}^{(S)}) = \frac{N_A N_B}{N_A + N_B} (\bar{y}_A - \bar{y}_B)^2.$$

- (a) Every contrast $\mathbf{d}$ that is zero outside A (or outside B) is orthogonal to $\mathbf{c}^{(S)}$ for the design.
- (b) Split the set of all g levels into two parts, split each part with more than one level into two, and continue until every part is a single level. The $g-1$ contrasts of the splits form a complete set of orthogonal contrasts, whatever the group sizes.

Proof. The estimate is $\sum_{k \in A} n_k \bar{y}_k / N_A - \sum_{k \in B} n_k \bar{y}_k / N_B = \bar{y}_A - \bar{y}_B$, and $v_c = \sum_{k \in A} n_k / N_A^2 + \sum_{k \in B} n_k / N_B^2 = 1/N_A + 1/N_B$, which gives the sum of squares. (a) If $\mathbf{d}$ is zero outside A , then $\sum_k c_k^{(S)} d_k / n_k = \sum_{k \in A} (n_k / N_A) d_k / n_k = N_A^{-1} \sum_{k \in A} d_k = 0$, since $\mathbf{d}$ is a contrast; the

case of B is the same. (b) A sequence of splits ending in single levels is a binary tree with g leaves, so it has $g - 1$ internal nodes, one split each. Take two different splits, of sets S and S' . By construction either S and S' are disjoint, in which case the two contrasts have disjoint supports and are orthogonal, or one of them, say S' , lies inside one part of the split of S , in which case (a) applies. So the $g - 1$ contrasts are pairwise orthogonal, and they form a complete set. ■

The weighting matters: comparing plain averages of the levels in A and B gives orthogonal contrasts, in general, only in a balanced layout, as Example (Splitting the education sum of squares) showed numerically. The Helmert contrasts of Exercise (B3), which compare each level with the average of those before it, are the special case of the proposition that splits off one level at a time; with the weights n_k/N_A they stay orthogonal for unequal group sizes.

Factorial treatments. Four treatments may be the combinations of two fertilizers, each present or absent: none, P only, K only, and both. The contrasts

$$\begin{aligned} \text{effect of } P &: (-1, 1, -1, 1), & \text{effect of } K &: (-1, -1, 1, 1), \\ \text{interaction} &: (1, -1, -1, 1) \end{aligned}$$

are orthogonal in a balanced layout. The third asks whether the effect of P depends on K . A one-way analysis with these contrasts is a two-factor analysis in disguise; Chapter 16 treats such layouts directly and defines interaction in general (Definition 16.2.1).

Polynomial trends. When the levels are values of a quantitative variable, such as a dose or a fertilizer rate, the orthogonal polynomial contrasts describe the shape of the response one degree at a time. For equally spaced levels and equal group sizes they are the integer vectors obtained from Gram–Schmidt applied to $1, s, s^2, \dots$ in the level scores (Section 8.7, Section 9.4). For five levels they are

$$\begin{aligned} \text{linear} & (-2, -1, 0, 1, 2), & \text{quadratic} & (2, -1, -2, -1, 2), \\ \text{cubic} & (-1, 2, 0, -2, 1), & \text{quartic} & (1, -4, 6, -4, 1). \end{aligned}$$

The degree- d contrast is orthogonal to every polynomial of lower degree in the score, so it vanishes whenever the means lie on a polynomial of degree less than d . With unequal spacing the Gram–Schmidt step uses the actual level values (Exercise (B2)), and with unequal group sizes it uses the weighted inner product $\sum_k n_k p_k q_k$ and the contrasts $c_k = n_k p_k$ (Example (Trend in ideology across party identification)).

15.4.3 Trend analysis of the nitrogen trial

Example. The shape of the nitrogen response

The five rates are equally spaced and the design is balanced, so the integer contrasts above apply. The four single-degree-of-freedom sums of squares, each tested against $s^2 = 0.2715$ on 25 degrees of freedom, are:

Component	Sum of squares	F	p	Share of SSB
linear	25.9121	95.42	5.1×10^{-10}	90.3%
quadratic	2.7904	10.28	0.0037	9.7%
cubic	0.0049	0.02	0.89	0.0%

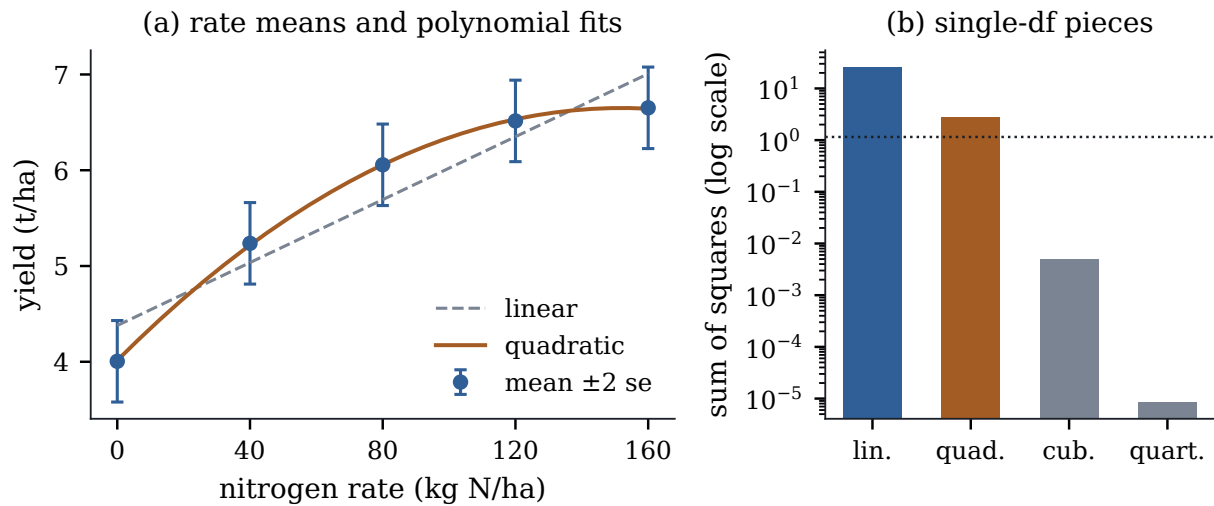


Figure 15.4.1: Trend analysis of the nitrogen trial. (a) Rate means with bars of two standard errors, and the least squares linear and quadratic curves in the rate fitted to all thirty plots. (b) The four orthogonal polynomial sums of squares (logarithmic scale); the dotted line is the value a single degree of freedom needs to be significant at level 0.05, $s^2 F_{0.05}(1, 25)$.

quartic	8.6×10^{-6}	3.2×10^{-5}	0.996	0.0%
rates (total)	28.7074	26.43		

The pieces add to the between-rates sum of squares of Example (The analysis of variance for the nitrogen trial), and F is the average of the four single-degree-of-freedom F values. Yield rises with the rate and the rise flattens: the quadratic component is clearly significant. The cubic and quartic components together test whether a quadratic in the rate describes the five means; their sum of squares 0.0049 on two degrees of freedom gives $F = 0.009$ and $p = 0.99$ (Figure 15.4.1).

```
import numpy as np
from scipy import stats

rates = np.array([0, 40, 80, 120, 160])           # kg N per hectare
g, m = len(rates), 6                               # 5 rates, 6 plots per rate
true_means = 6.8 - 2.7 * np.exp(-0.018 * rates)    # unknown to the analyst
rng = np.random.default_rng(1515)
group = np.repeat(np.arange(g), m)                 # plot i received rate group[i]
y = np.round(true_means[group] + rng.normal(0, 0.5, g * m), 2) # yield, t/ha
n = len(y)

means = np.array([y[group == k].mean() for k in range(g)])
nu = n - g
s2 = np.sum((y - means[group]) ** 2) / nu
ss_between = m * np.sum((means - means.mean()) ** 2)

C_poly = np.array([[[-2, -1, 0, 1, 2],           # linear
                    [2, -1, -2, -1, 2],          # quadratic
                    [-1, 2, 0, -2, 1],           # cubic
                    [1, -4, 6, -4, 1]], float)     # quartic

def contrast_ss(c):
    return (c @ means) ** 2 / (c @ c / m)          # balanced: sum c_k^2 / n_k = c'
```



```
[0, -1, 3, -3, 1]], float)          # cubic among them
ss_ctrl = np.array([contrast_ss(c) for c in C_ctrl])
print("second set:", ss_ctrl.round(4), " sum =", round(ss_ctrl.sum(), 4))
```

[Run this code online](#)

The two decompositions tell different but compatible stories. In the first, curvature is significant; in the second, the curvature among the fertilized rates is not, because most of the bend is the jump from no nitrogen to 40 kg, which the second set assigns to the control contrast. Computing several decompositions and reporting the one whose pieces look best is the data snooping of [Section 13.6](#).

15.4.5 Exercises

15.4.5.1 A. Check your understanding

Exercise. A1

Check that the four polynomial contrasts for five levels are pairwise orthogonal. Show that the quadratic contrast vanishes when $\mu_k = a + bk$, and the cubic contrast vanishes when $\mu_k = a + bk + ck^2$.

Exercise. A2

For the four fertilizer combinations of “Factorial treatments”, verify that the three contrasts are orthogonal when every group has the same size, and that the P and K contrasts are not orthogonal for group sizes 3, 3, 3, 5 (in the order none, P , K , both). What does the lack of orthogonality mean for the estimates?

Solution. With equal sizes, orthogonality is the plain inner product: $(-1, 1, -1, 1) \cdot (-1, -1, 1, 1) = 1 - 1 - 1 + 1 = 0$, and similarly for the other two pairs. With sizes 3, 3, 3, 5, $\sum_k c_k d_k / n_k = \frac{1}{3} - \frac{1}{3} - \frac{1}{3} + \frac{1}{5} = -\frac{2}{15} \neq 0$. By Proposition 8.7.2(c) the two estimates are then correlated, with covariance $-\frac{2}{15}\sigma^2$, and their sums of squares do not add to the part of SSB that the two contrasts span.

15.4.5.2 B. Practice

Exercise. B1

Show that the two planned contrasts of Example (Two planned contrasts), $(-1, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ and $(0, 0, -1, 0, 1)$, are orthogonal. Find two further contrasts that complete them to a complete orthogonal set, and say what they compare.

Solution. The design is balanced, and $-1 \cdot 0 + \frac{1}{4}(0 - 1 + 0 + 1) = 0$. A contrast $\mathbf{e}$ orthogonal to the first satisfies $-e_1 + \frac{1}{4}(e_2 + e_3 + e_4 + e_5) = 0$, and since $e_2 + e_3 + e_4 + e_5 = -e_1$ this forces $e_1 = 0$. Orthogonality to the second forces $e_3 = e_5$. So $\mathbf{e} = (0, a, b, c, b)$ with $a + 2b + c = 0$. Two orthogonal choices are $(0, 1, 0, -1, 0)$, comparing 40 with 120, and $(0, 1, -1, 1, -1)$, comparing 40 and 120 together with 80 and 160 together. Their inner

product is $1 - 1 = 0$. The completion is valid but not very natural: completing a planned set is optional.

Exercise. B2

Five plots at each of four rates 0, 1, 2 and 4 (in units of 50 kg N/ha) are to be analysed for trend. Show that the linear and quadratic orthogonal polynomial contrasts are proportional to $(-7, -3, 1, 9)$ and $(7, -4, -8, 5)$. Check that the quadratic contrast vanishes when the means are linear in the rate.

Solution. The rates have mean $7/4$, so the linear contrast is proportional to $4(s_k - 7/4) = (-7, -3, 1, 9)$. For the quadratic, centre $s_k^2 = (0, 1, 4, 16)$ at its mean $21/4$ and multiply by four: $(-21, -17, -5, 43)$. Its inner product with $(-7, -3, 1, 9)$ is 580, and the linear vector has squared length 140, so subtract $\frac{580}{140}(-7, -3, 1, 9)$ to get $(8, -\frac{32}{7}, -\frac{64}{7}, \frac{40}{7})$, which is proportional to $(7, -4, -8, 5)$. Its entries sum to zero and its inner product with the linear contrast is $-49 + 12 - 8 + 45 = 0$. For means $a + bs_k$ the quadratic contrast gives $b(0 - 4 - 16 + 20) = 0$.

Exercise. B3

Four equally spaced levels have group sizes 2, 4, 4, 2. Show that the classical linear contrast $(-3, -1, 1, 3)$ is orthogonal for this design to the quadratic $(1, -1, -1, 1)$ but not to the cubic $(-1, 3, -3, 1)$. Explain the first fact by symmetry.

Solution. For linear and quadratic, $\sum_k c_k d_k / n_k = -\frac{3}{2} + \frac{1}{4} - \frac{1}{4} + \frac{3}{2} = 0$. For linear and cubic, it is $\frac{3}{2} - \frac{3}{4} - \frac{3}{4} + \frac{3}{2} = \frac{3}{2} \neq 0$. The sizes are symmetric about the middle of the scale. The linear contrast is odd under the reflection $k \mapsto 5 - k$, the quadratic is even, and the weights $1/n_k$ are even, so the weighted inner product of an odd and an even vector vanishes. Linear and cubic are both odd, and nothing forces their weighted inner product to vanish.

Exercise. B4

Use the formula of [Proposition 15.4.2](#) to recompute the sum of squares of the control against the fertilized rates in the nitrogen trial from the rate means, and compare it with the value 21.3701 in the text.

Solution. Here $N_A = 24$, $N_B = 6$, $\bar{y}_A = \frac{1}{4}(5.237 + 6.057 + 6.515 + 6.652) = 6.11525$ and $\bar{y}_B = 4.005$. The sum of squares is $\frac{24 \cdot 6}{30}(6.11525 - 4.005)^2 = 4.8 \times 2.11025^2 \approx 21.375$. The small difference from 21.3701 comes from using the rounded rate means.

15.4.5.3 C. Going deeper

Exercise. C1

Let $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_{g-1}]$ and $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_{g-1}]$ be two orthonormal bases of the treatment space. Show that $\mathbf{V} = \mathbf{U}\mathbf{Q}$ for an orthogonal $(g-1) \times (g-1)$ matrix $\mathbf{Q}$, that the vectors of squared coordinates of $\mathbf{y}$ in the two bases have the same sum SSB, and that the individual squared coordinates generally differ. More generally, show that for any symmetric idempotent $\mathbf{A}$ of rank q and $\mathbf{Y} \sim N_n(\boldsymbol{\theta}, \sigma^2 \mathbf{I})$, $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ splits into q independent scaled noncentral $\chi^2(1)$ variables, one for each vector of an orthonormal basis of $\mathcal{C}(\mathbf{A})$.

Solution. Both $\mathbf{U}\mathbf{U}^\top$ and $\mathbf{V}\mathbf{V}^\top$ equal $\mathbf{P}_\mathcal{T}$, the unique projection onto $\mathcal{T}$. Put $\mathbf{Q} = \mathbf{U}^\top \mathbf{V}$. Then $\mathbf{U}\mathbf{Q} = \mathbf{P}_\mathcal{T} \mathbf{V} = \mathbf{V}$ and $\mathbf{Q}^\top \mathbf{Q} = \mathbf{V}^\top \mathbf{P}_\mathcal{T} \mathbf{V} = \mathbf{V}^\top \mathbf{V} = \mathbf{I}$. The coordinate vectors satisfy $\mathbf{V}^\top \mathbf{y} = \mathbf{Q}^\top \mathbf{U}^\top \mathbf{y}$, and an orthogonal matrix preserves length, so both have squared length $\mathbf{y}^\top \mathbf{P}_\mathcal{T} \mathbf{y} = \text{SSB}$. The individual coordinates are mixed by $\mathbf{Q}$ and generally change. For the general statement, write $\mathbf{A} = \sum_{i=1}^q \mathbf{w}_i \mathbf{w}_i^\top$ with $\mathbf{w}_1, \dots, \mathbf{w}_q$ an orthonormal basis of $\mathcal{C}(\mathbf{A})$ (Proposition 6.3.2). The rank-one projections $\mathbf{w}_i \mathbf{w}_i^\top$ are mutually orthogonal, so by Theorem 4.4.4 the terms $(\mathbf{w}_i^\top \mathbf{Y})^2$ are independent, and $(\mathbf{w}_i^\top \mathbf{Y})^2 / \sigma^2 \sim \chi^2(1, (\mathbf{w}_i^\top \boldsymbol{\theta})^2 / \sigma^2)$. Eigenvectors of $\mathbf{A}$ for the eigenvalue one are one such basis.

Exercise. C2

For g equally spaced levels with equal group sizes, show that the sum of the sums of squares of the polynomial contrasts of degrees $d+1, \dots, g-1$ is the lack-of-fit sum of squares of the polynomial regression of degree $d < g-1$ on the level score, and that the sum for degrees $1, \dots, d$ is that regression's sum of squares after the mean.

Solution. Let $\mathcal{P}_d \subseteq \mathcal{C}(\mathbf{Z})$ be the mean space of the degree- d regression, spanned by the vectors whose entry in group k is s_k^i , $i = 0, \dots, d$ (Lemma 14.6.1). The contrast vector of the degree- i orthogonal polynomial p_i has entry $p_i(s_k)/m$ in group k , and in a balanced layout orthogonality of these vectors is orthogonality of the p_i in the plain inner product. By construction $p_0, \dots, p_d$ span the polynomials of degree at most d on the scores, so the vectors of degrees $1, \dots, d$ span $\mathcal{P}_d \cap \mathbf{1}^\perp$, and those of degrees $d+1, \dots, g-1$ span $\mathcal{C}(\mathbf{Z}) \cap \mathcal{P}_d^\perp$. The projection onto the first space gives the regression sum of squares after the mean (Theorem 6.5.1), and the projection onto the second gives $\|(\mathbf{M} - \mathbf{P}_{\mathcal{P}_d})\mathbf{y}\|^2$, the lack-of-fit sum of squares of Theorem 14.6.2. Each projection is the sum of the projections onto the lines, by Theorem 6.3.7.